Software Engineering

Subject: Computer Science | NoteShare Library

SOFTWARE ENGINEERING

1. SDLC
Requirements -> Design -> Implementation
-> Testing -> Deployment -> Maintenance

2. MODELS
Waterfall (linear)
Incremental
Spiral (risk-driven)
Agile (iterative)
Scrum, Kanban

3. REQUIREMENTS
Functional
Non-functional
Use cases, User stories

4. DESIGN
SOLID principles:
Single Responsibility
Open/Closed
Liskov Substitution
Interface Segregation
Dependency Inversion

Design patterns:
Creational: Singleton, Factory
Structural: Adapter, Decorator
Behavioral: Observer, Strategy

5. TESTING
Unit, Integration, System
Black-box, White-box
TDD, BDD
JUnit, pytest

6. VERSION CONTROL
Git: init, clone, add, commit, push, pull
Branching: feature, main
Merge, Rebase, PR

7. CI/CD
Jenkins, GitHub Actions
Docker, Kubernetes